Vectors between points

The displacement from A to B is the difference of the two position vectors measured from the origin,

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}.$$

Reversing the order of the points reverses the vector:

$$\overrightarrow{AB} = -\overrightarrow{BA}.$$

Two triangles with equal sides are congruent, which we write

$$\triangle ABC \cong \triangle DEF.$$

Updates and logic

A loop that counts down starts with the assignment

$$x \leftarrow -1$$

and a premise set that entails a formula is written

$$\Gamma \models \varphi.$$

Work and torque

The work done by a constant force over a displacement is a dot product,

$$W = \vec{F} \cdot \vec{d},$$

while the torque about the origin is a cross product,

$$\vec{\tau} = \vec{r} \times \vec{F}.$$